

LABORATORY OF AUTOMATION SYSTEMS

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Today Goal:

- Play with our PI
 - Omega vs Omega Filtered
 - effect of speed filtering on stability
 - Check loops time constant
- Reference Trajectory
- FeedForward with reference trajectory

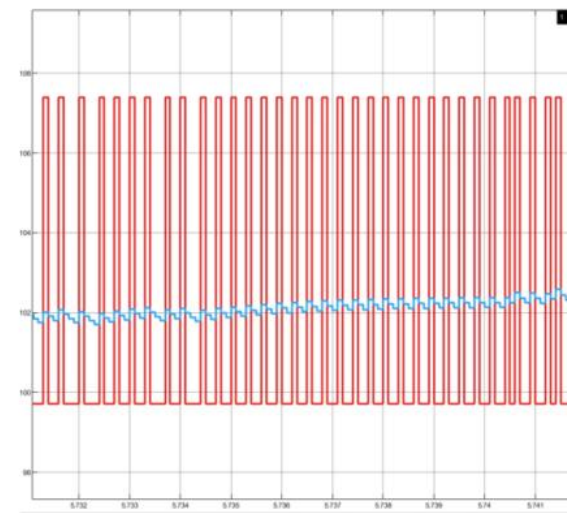
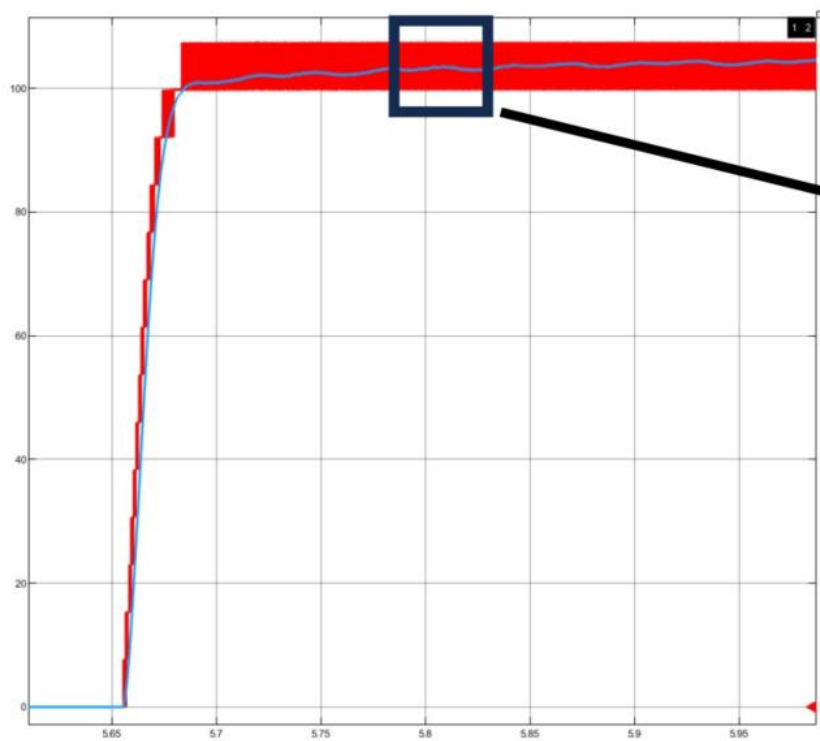
You can open the model made in the last lesson

If your control is OK, you can increase:

- current limit (Model Saturation + Power Supply) to **6 Ampere [V/C]** **6.5 Ampere [OVP/OVC]**
- Voltage (Model Saturation + Power Supply) to **32V [V/C]** **35V Ampere [OVP/OVC]**

- Watch Omega and OmegaFilt

ALCUNI ESEMPI DI TRACING: **OmegaMis**
OmegaFilt

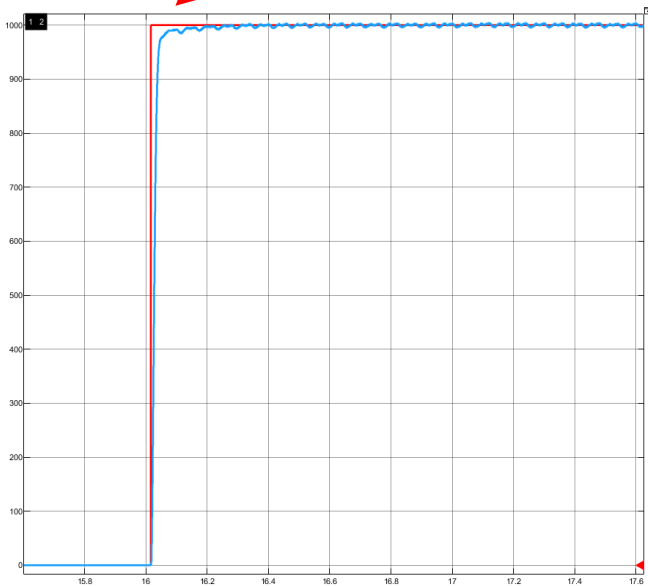


% Omega filter
TaoFilterOmega=0.002;

- test different value of TaoOmegaFilt.

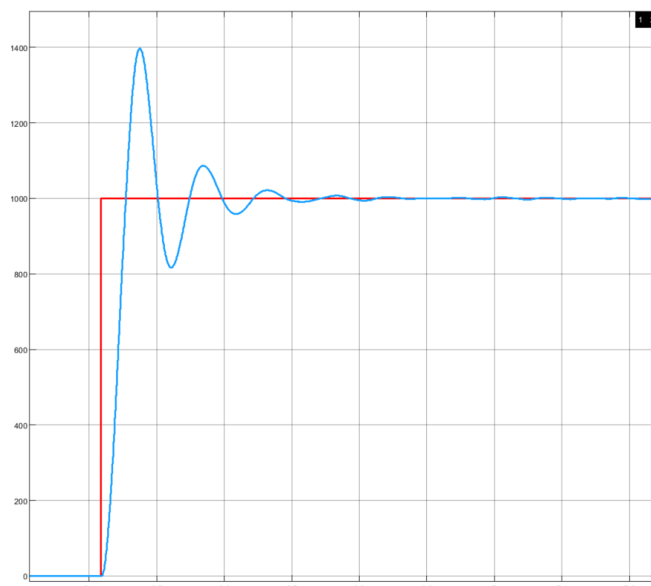
You can have some overshoot

We will remove it with motor parameter esimation



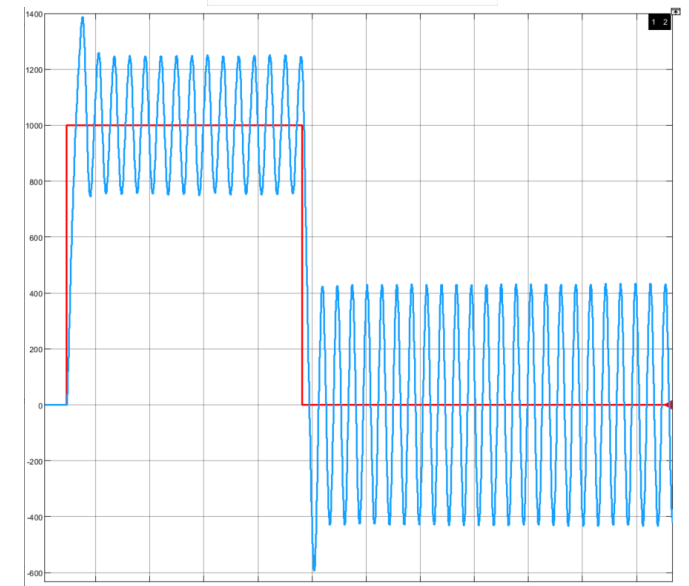
% Omega filter

TaoFilterOmega=0.002;



% Omega filter

TaoFilterOmega=0.02;



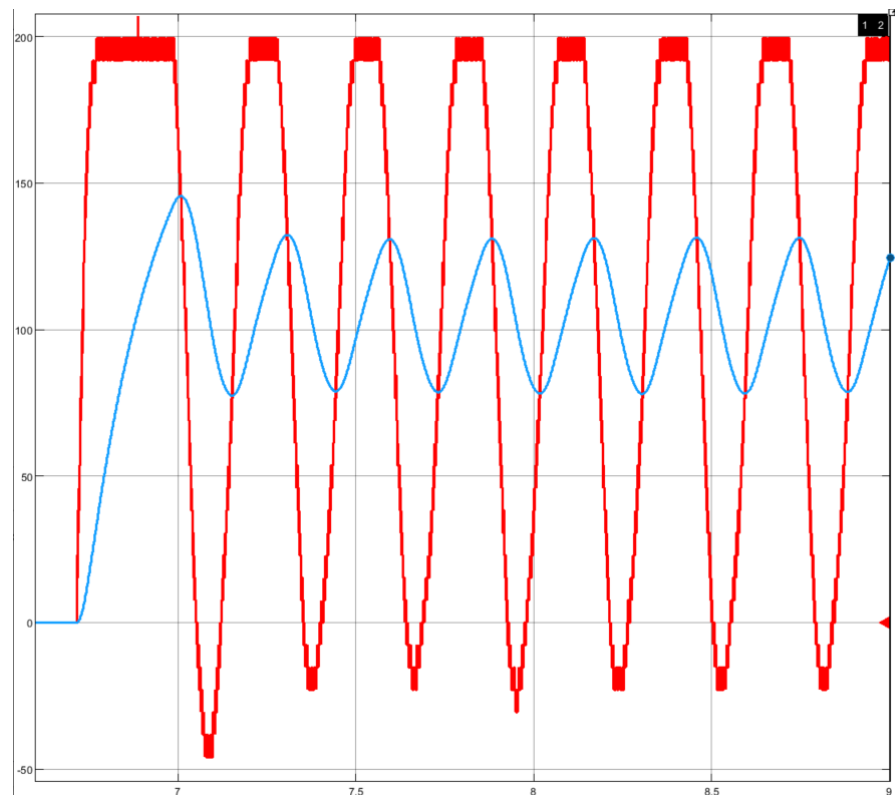
% Omega filter

TaoFilterOmega=0.2;



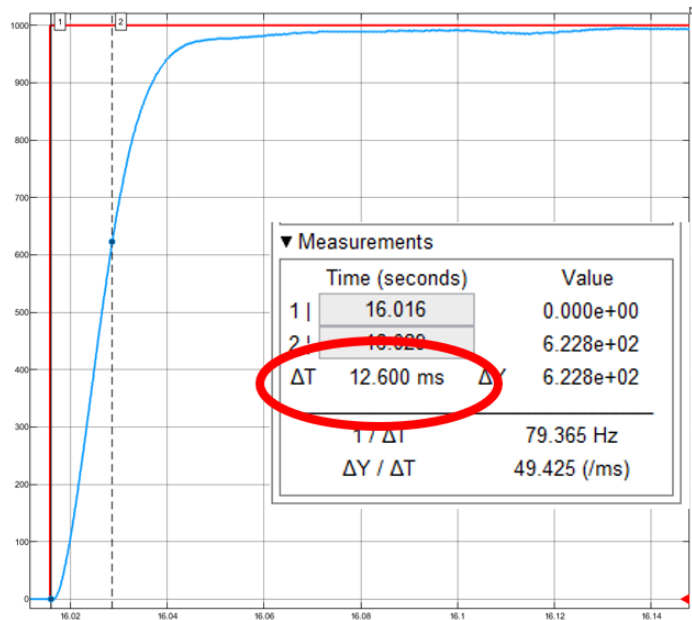
- test different value of $TaoOmegaFilt$.

OmegaMis vs OmegaFilt



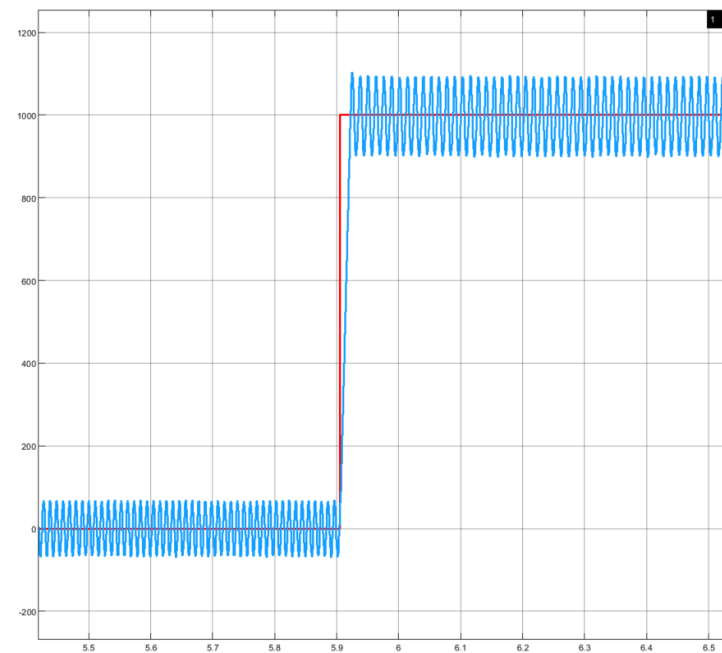
- Check speed loop Time Constant and test the effects of changing T_{iw} :
For example: $T_{aow}/10$

From last lesson



%Desired T_{aow}

$T_{aow}=10e-3;$



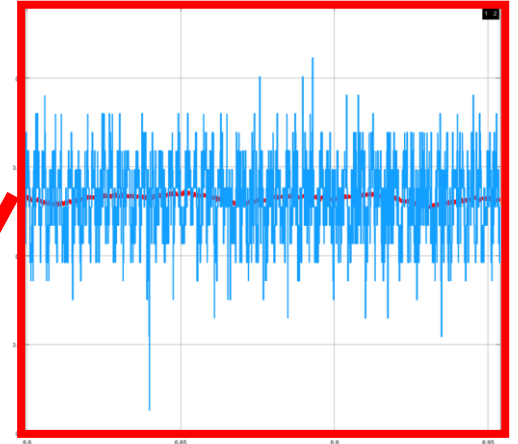
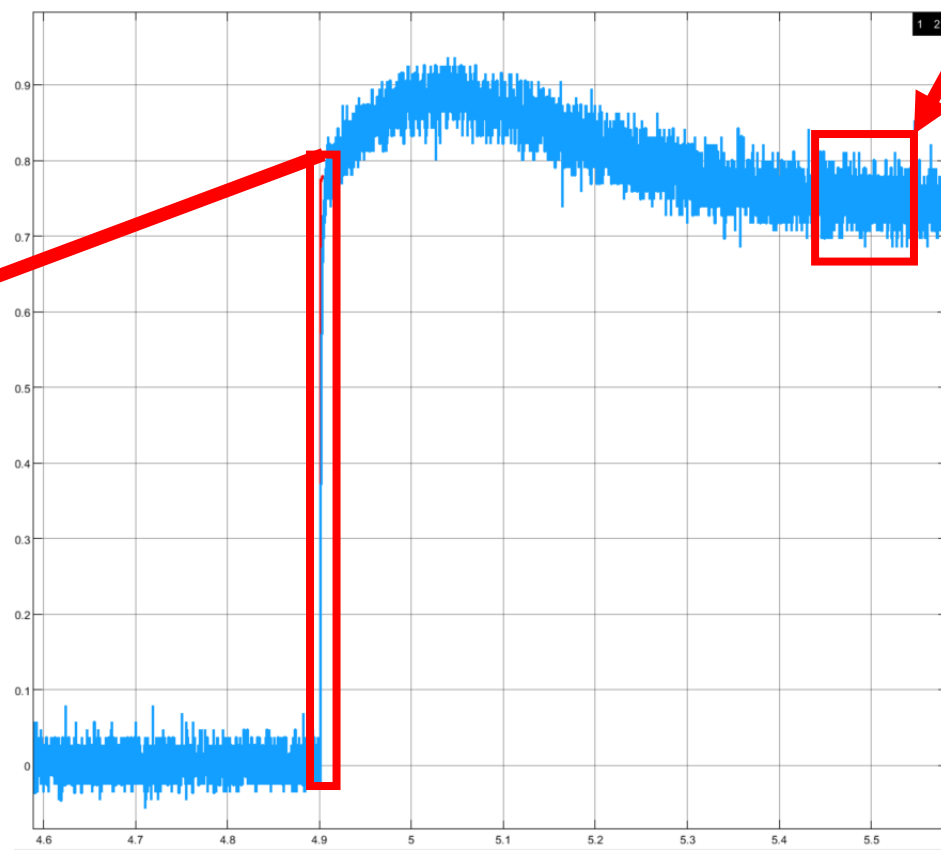
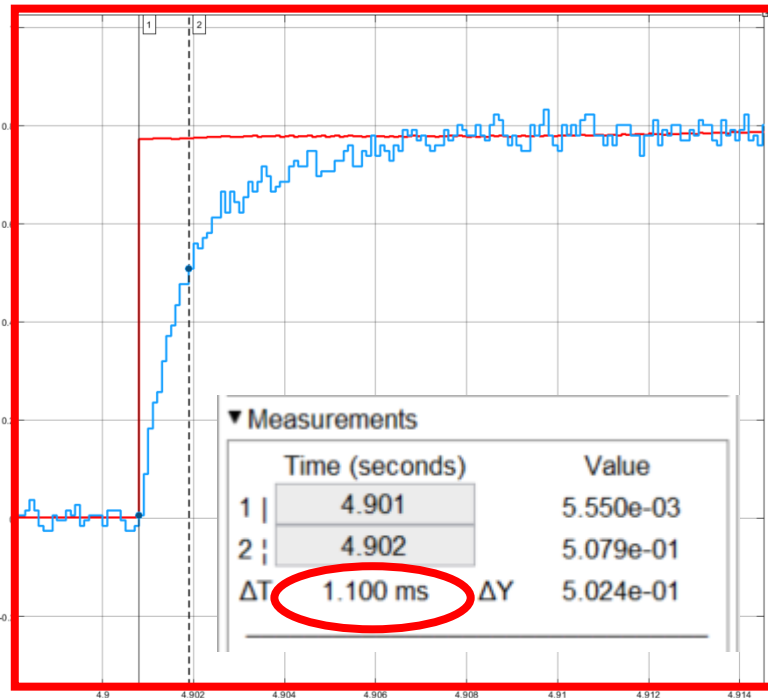
%Desired T_{aow}

$T_{aow}=1e-3;$

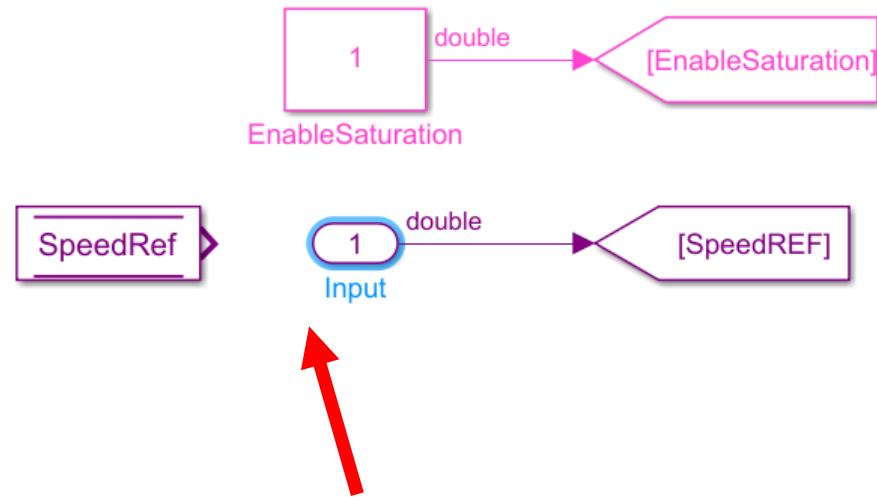


- Check your current control

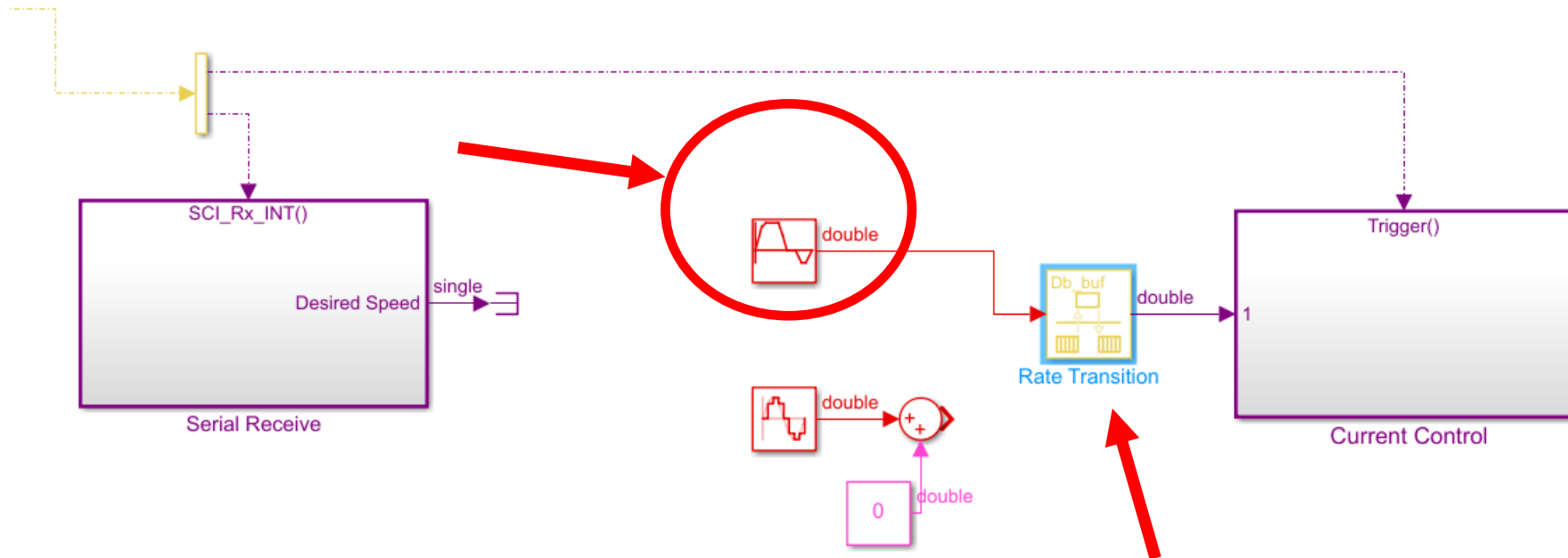
ISetPoint-Imes



GIVE A REFERENCE TRAJECTORY INSTEAD A FIXED DESIRED



Disconnect SpeedRef and put an input port

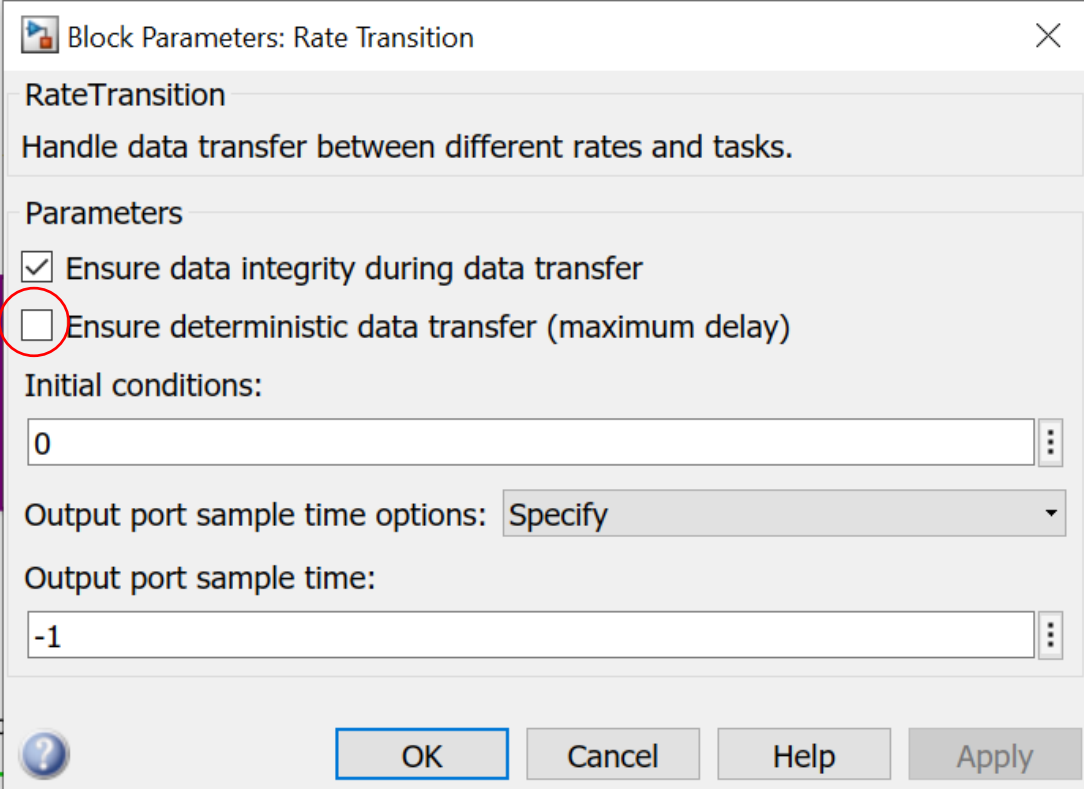


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In the upper level you can copy and paste from **ReferenceTrajectory.xls**

Test both trajectory

Rate transition:



Block Parameters: Rate Transition

RateTransition
Handle data transfer between different rates and tasks.

Parameters

- ☒ Ensure data integrity during data transfer
- ☐ Ensure deterministic data transfer (maximum delay)

Initial conditions:

0

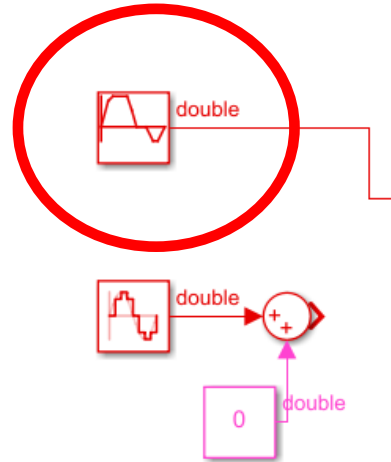
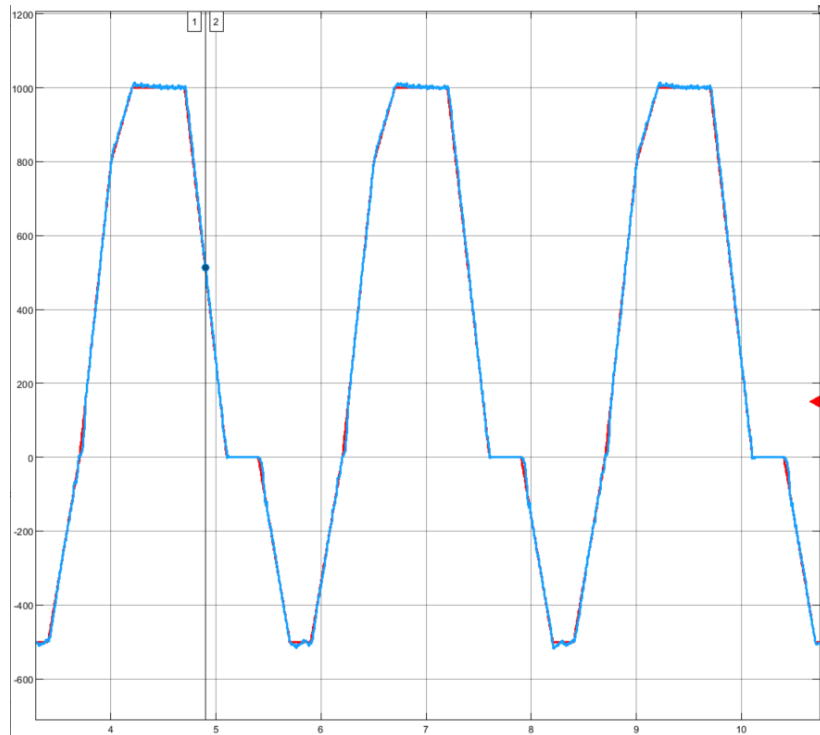
Output port sample time options: Specify

Output port sample time:

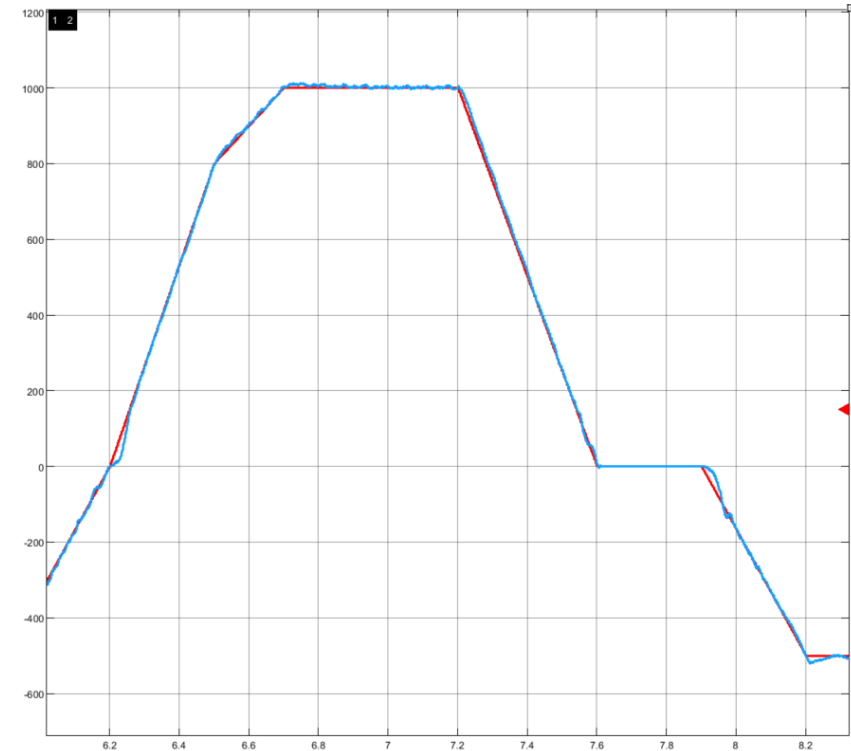
-1

OK Cancel Help Apply

OmegaSetPoint-OmegaMis



OmegaSetPoint-OmegaMis



If you want to compare data with small tracking error, it is best if you show error signal



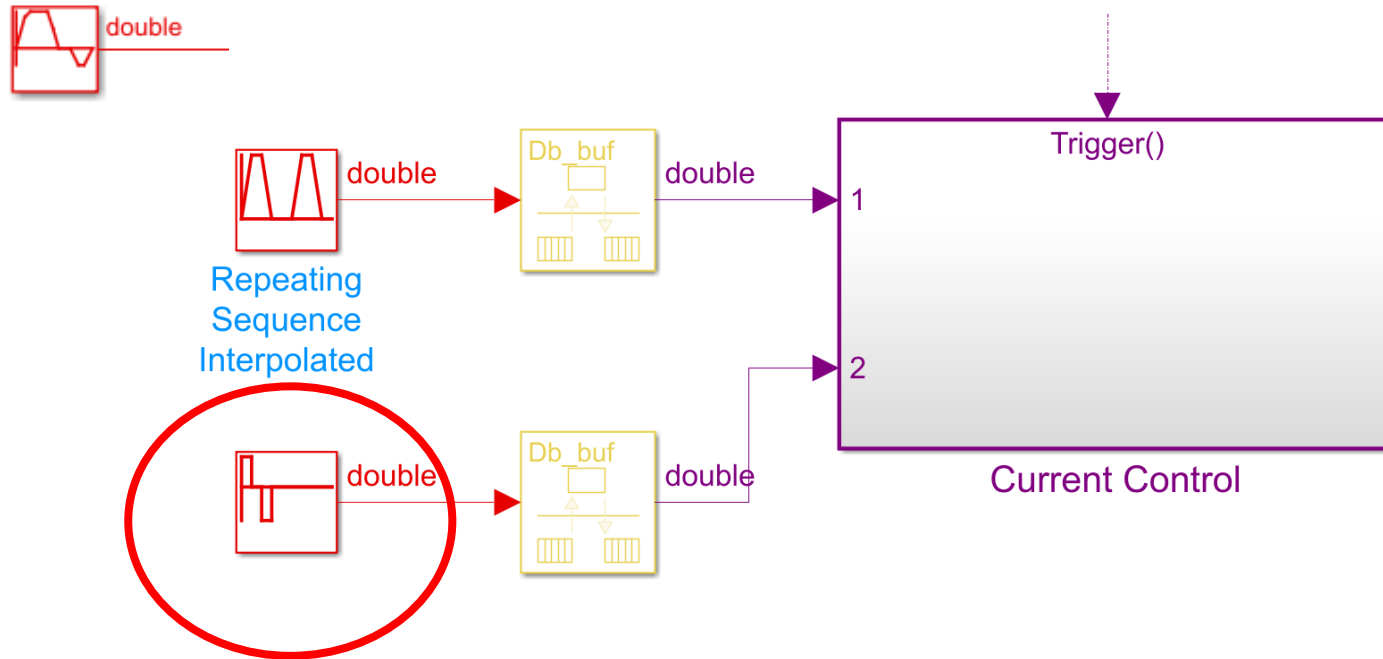
REFERENCE TRAJECTORY – FEEDFORWARD (Prof. Melchiorri Slide)

- **FF - Current**

$$i_{ff} = \frac{J}{K_m} * \ddot{\theta}_d$$

From the velocity trajectory we can derive the required acceleration

Trajectory in file: **FeedForwardReferenceTrajectory.xls**

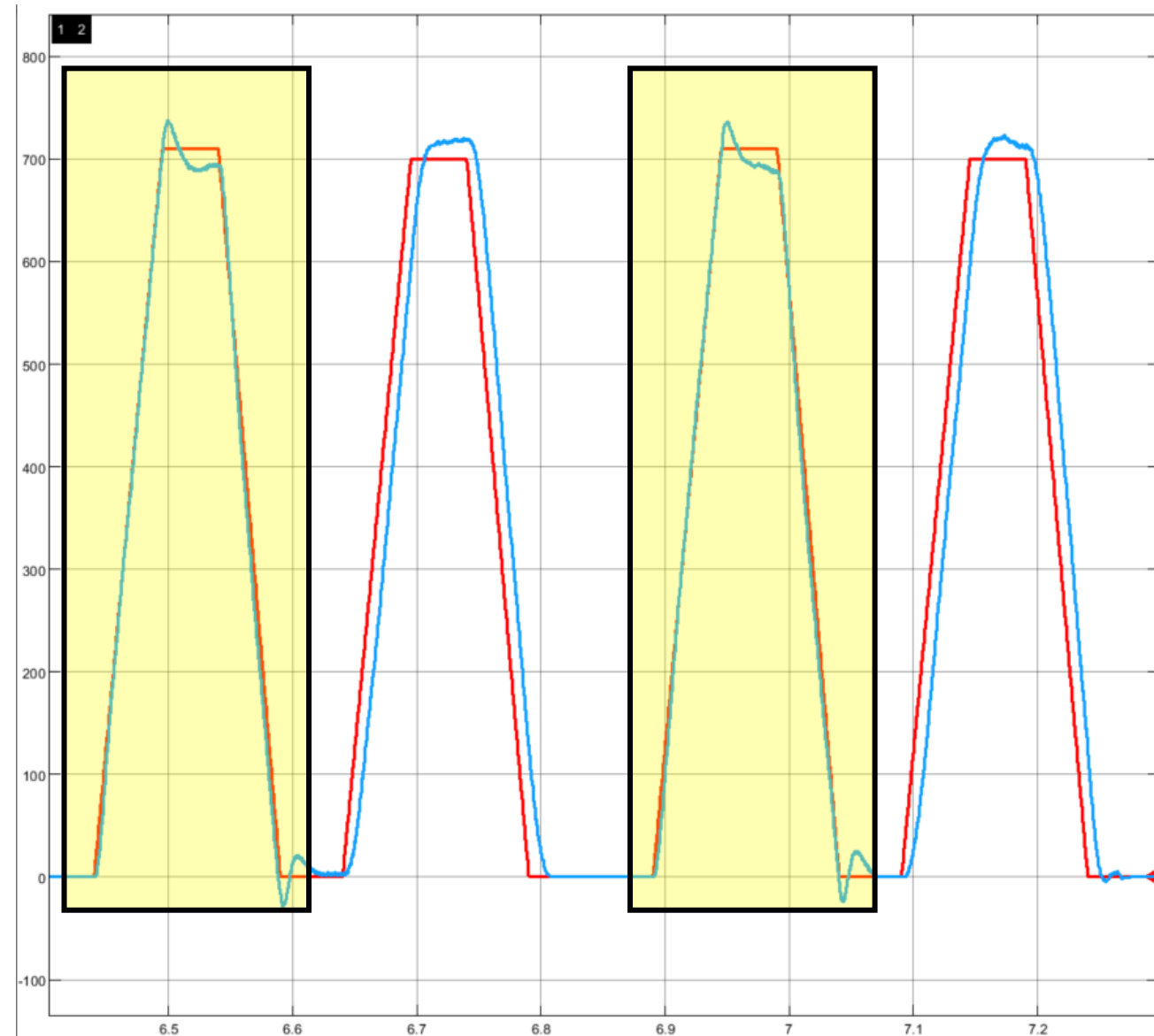


You can get the result in different way

To better see the effect of FF we want to implement it only every other trapezoid.

Better, We can get data and plot in Matlab...

Error Plot can help to show difference



PI END